

Careers
NLM
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give me the gaps of these projects and also the reference links of these and a brief of every projects

Beautiful. You've brought three heavyweights to the table:

1. **Speech Emotion Recognition: A Comprehensive Survey (2023)**
2. **Improved SER with Mel Frequency Magnitude Coefficient (MFMC) – Applied Acoustics, 2021**
3. **RAVDESS Dataset Paper – PLOS ONE, 2018**

Let's dissect them like researchers, not just readers.

I'll give you for each:

- 📌 Brief summary
- ⚠️ Research gaps
- 🔗 Reference link
- 💡 How you can improve or extend it

1 Speech Emotion Recognition: A Comprehensive Survey (2023)

📄 **Title:** Speech Emotion Recognition: A Comprehensive Survey

📘 **Journal:** Wireless Personal Communications (2023)

🔗 **DOI:** <https://doi.org/10.1007/s11277-023-10244-3> ↗

📌 Brief

This is a **survey paper**, not a model-building paper.

It reviews:

- SER databases
- Feature extraction techniques (MFCC, prosodic, spectral)

- Classifiers (SVM, CNN, RNN, hybrid models)
- Multiclassifier systems
- Challenges in SER
- Future directions

Think of it as a satellite view of the whole SER galaxy 🌌

⚠️ Gaps in This Survey

1. ❌ No experimental validation
It does not propose a new model.
 2. ❌ Limited focus on self-supervised models
Modern transformers like wav2vec 2.0 are underexplored.
 3. ❌ No large-scale cross-dataset evaluation
Most reviewed studies use small acted datasets.
 4. ❌ Limited real-world robustness discussion
Noise, spontaneous emotion, real call-center data are not deeply analyzed.
 5. ❌ Weak discussion on multilingual SER
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💡 How You Can Improve

You can:

- Build a transformer-based SER system (wav2vec 2.0 / HuBERT)
- Evaluate across 3–4 datasets
- Add noise robustness testing
- Do cross-corpus evaluation
- Compare dimensional emotion models (valence-arousal)

That would be a strong journal paper.

2 Improved Speech Emotion Recognition with MFMC (2021)

- 📄 **Title:** Improved speech emotion recognition with Mel frequency magnitude coefficient
- 📘 **Journal:** Applied Acoustics (Elsevier), 2021
- 🔗 **DOI:** <https://doi.org/10.1016/j.apacoust.2021.108046> ↗

Brief

They modified MFCC extraction:

Traditional MFCC:

- Uses power spectrum
- Uses DCT

Their MFMC:

- Uses magnitude spectrum instead of energy
- Removes DCT

Classifier used:

- Multiclass SVM

Datasets used:

- Berlin
- RAVDESS
- SAVEE
- EMOVO
- eINTERFACE
- Urdu database

Reported accuracy (example):

- Berlin: 81.50%
- RAVDESS: 64.31%
- Urdu: 95.25%

Gaps in MFMC Paper

1.  Uses classical ML (SVM only)
2.  No deep learning comparison (CNN, LSTM, Transformer)
3.  No cross-corpus testing
4.  No speaker-independent evaluation discussion
5.  Small datasets only
6.  No noise robustness evaluation
7.  No real-time deployment experiment

This is a feature-engineering era paper. Today the field has moved toward representation learning.

How You Can Improve

You can:

- Replace SVM with CNN-BiLSTM
- Combine MFMC + wav2vec embeddings
- Add SpecAugment
- Perform cross-corpus training
- Evaluate speaker-independent splits
- Compare MFMC vs Log-Mel spectrogram vs pretrained embeddings

That would significantly outperform their 64–81% results.

RAVDESS Dataset Paper (2018)

 **Title:** The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS)

 **Journal:** PLOS ONE (2018)

 **Dataset link:** <https://doi.org/10.5281/zenodo.1188976> ↗

 **Paper:** <https://doi.org/10.1371/journal.pone.0196391> ↗

Brief

This is a **dataset creation and validation paper**.

Features:

- 7356 recordings
- 24 actors
- Speech + Song
- 8 emotions (speech)
- 6 emotions (song)
- 3 modalities:
 - Audio-Visual
 - Video-only

- Audio-only

They validated:

- Emotional accuracy
- Intensity
- Genuineness
- Test-retest reliability

Recognition rates:

- Audio-visual $\approx 80\%$
- Video-only $\approx 75\%$
- Audio-only $\approx 60\%$

Gaps in RAVDESS

1.  Acted emotions only
Not spontaneous
2.  Only 2 fixed sentences
Low lexical diversity
3.  Small speaker pool (24 actors)
4.  English only
5.  Controlled lab environment
No background noise
6.  Class imbalance in intensity levels
7.  No dimensional (valence-arousal) labels

How You Can Improve

You can:

- Combine RAVDESS with:
 - MSP-IMPROV
 - IEMOCAP
 - CREMA-D
 - Perform cross-corpus validation
 - Add real noisy data

- Use wav2vec + multimodal fusion
- Add valence-arousal regression



Big Combined Research Gap Across All Three

Here's the bigger picture:

Area	Gap
Dataset	Mostly acted
Size	Small datasets
Real-world	Not evaluated
Cross-corpus	Rare
Self-supervised models	Underused
Noise robustness	Weak
Multilingual SER	Limited
Emotion dimensions	Mostly categorical



If You Want a High-Accuracy Modern Project

Here's a strong research direction:



Project Idea:

"Cross-Corpus Transformer-Based Speech Emotion Recognition Using Self-Supervised Learning"

Use:

- wav2vec 2.0
- HuBERT
- Data augmentation
- Multi-dataset training

Datasets:

- RAVDESS
- IEMOCAP
- MSP-IMPROV
- CREMA-D
- Berlin

Add:

- Cross-dataset testing
- Speaker-independent splits
- Noise injection experiments

That is PhD-level strong.

If you want, I can now:

- Design a full research proposal
- Provide dataset download links
- Give a high-accuracy architecture diagram
- Provide code structure in PyTorch
- Help you frame a publishable research paper

Tell me what level you're aiming for — final year project, M.Tech, or journal publication?

